The paper "Which Model to Use for Cortical Spiking Neurons?" (2004) by Eugene Izhikevich is landmark guide in computational neuroscience.

① Trade-off :

The Izhikevich explains that when you build a brain simulation, you are always fighting a battle betw

Fidelity
- How much the model acts like a real cell

Efficiency
How much thousands of these neurons your computer can handle at once.

- He compares several famous model, ranging from the very simple [Integrate-and-Fire] to the very complex [Hodgkin-Huxley], and places his own model right in the "Sweet Spot".

② Neuron's type. (how neuron reacts to a steady input of current)

These neurons fire as long as they are being stimulated. They are the "workhorses" of brain.

CONTINUOUS RESPONDERS (BASIC SPIKING)

1. Tonic spiking: The most basic type As long the input is "on", the neuron fires at a steady frequency.
2. Phasic Spiking: The neuron fires a single spike when the input starts, then goes silent. It acts like a change detector.
3. Tonic Bursting: Instead of single spikes, it fires "clusters" of spikes repeatedly.
4. Phasic Bursting: It fires one cluster of spikes at the very beginning of the stimulus and then stops.

Category 2 : Frequency & Timing Specialists.

These neurons care about when and/or how fast the signal arrives.

5. Accommodation: The neuron is sensitive to the rate of change. If you increase the current slowly, it won't fire. It only fires if the current hits it quickly.

6. Integrator: The neuron adds up small input over time. Once the "bucket" is full, it fires.

7. ~~Int~~ Resonator: The neuron "prefers" a specific frequency. It will only fires if the input pulses at a rhythm that matches its own internal clock.

8. Input Facilitation: The neuron is "primed". If you give it a small pulses, it doesn't fire, but it becomes much easier to trigger with a second pulse shortly after.

Category 3 : The "Post Inhibition" Responders.

These neurons are unique because they fire when they are released from being suppressed. Imagine a spring being held down, when ~~they to~~ you let go, it jumps.

9. Rebound Spike: The neuron is quiet while being suppressed, but it fires one "rebound" spike the moment the suppression stops.

10. Rebound Burst: Same as above, but it fires a "cluster" (burst) of spikes when released.

11. Threshold Variability: The voltage needed to fire changes based on the neuron's voltage / recent history. Sometimes a small signal triggers it, sometimes it needs a huge one.

12. Inhibition - Induced spiking: Paradoxically, the neuron only starts firing because it is being suppressed in a specific rhythmic way.

Category 4: Complex & Erratic Behaviors.

- These are the "irregular" patterns found in specific parts of cortex or thalamus.

13. Class 1 Excitable: The neuron can fire at very low frequencies. As you increase current, the frequency grows smoothly from zero

14. Class 2 Excitable: The neuron has a "minimum speed". It is either silent or firing at a high frequency; it can't do "slow".

15. Bistability: The neuron can be in two states: either "resting" or "continuously firing". A tiny pulse can toggle it betw the two.

16. Depolarizing After-potential: After a spike, the voltage stays high for a while, making it very likely to fire again soon.

17. Subthreshold Oscillations: The voltage wiggles up and down like a wave without actually firing. This helps the brain synchronize different areas.

18. Mixed Mode: A combination behavior- it starts with a burst of spikes and then settles into steady, single spikes.

19. Chattering: High- frequency bursts separated by short quiet periods. These are the "talkative" neurons of visual cortex.

20. Spike Frequency Adaptation: The neuron starts firing fast, but then "gets tired" and slows down even though the input hasn't changed.

Explanation:

The brain isn't just made of 'on' and 'off' switch. Some neurons are like alarm clocks (Tonic), some are like motion sensors (Phasic) and some are like musicians that only play when they hear the right rhythm (Resonators). Izhikevich's math is so powerful because it play all 20 of these 'instruments' using the same four knobs (a, b, c, d).